Linux Fundamentals

What is Linux, and why is it widely used?

.linus is an operating system and it is widely used because it is free

and open source

What is the difference between the Linux kernel and a Linux distribution?

.the linux kernel is the central of the operating system and the Linux distribution packages the Linux kernel

Name three popular Linux distributions and their primary use cases.

.kali

.ubuntu

.debian

Linux Filesystem

What is the purpose of the root directory (/)?

The root directory is the directory of the entire Linux filesystem

What is the difference between /home and /root?

./home contains the personal directories of the normal users

./root the personal home directory of the root user

Which directory stores system configuration files?

./etc

Which directory stores most log files?

./var/log

Navigation and File Management

Which command displays your current directory?

.pwd

Which command changes directories?

.cd

Which command lists the contents of a directory?

.ls

How do you create a new directory?

.mkdir

How do you create an empty file?

.touch

Which commands are used to copy, move, and delete files?

.cp , mv , rm

Searching

What is the difference between find and grep?

.find is for searching for files/directories, grep searches inside files

Which command searches for files?

.find

Which command searches inside files?

.grep

How do wildcards help when searching?

.Wildcards allow you to match multiple files or patterns at once

Basic Commands

What does the echo command do?

.for terminal display

What is the purpose of the history command?

.it displays the commands you previously executed in the terminal

What does the pipe (|) operator do?

.the pipe sends the output of one command as the input of another command

What is the difference between > and >>?

. > :Creates the file or replaces its existing content

. >> :Adds to the end without deleting existing content

How can you access a command's manual page?

. man "command"

Users and Groups

What is the difference between a regular user and the root user?

.Regular user : limited permissions

.root user : high privileges

What is a group?

.a collection of users

What does the sudo command do?

.temporarily perform a command with administrator privileges

Which command displays information about the current user?

.id

Permissions

What do the r, w, and x permissions represent?

.r:read,w:write,x:execute

What is the difference between file ownership and group ownership?

.User/owner : the specific user who owns the file.

.Group : the group associated with the file.

Which command changes file permissions?

.chmod

Which command changes file ownership?

.chown

Important System Files

What information is stored in /etc/passwd?

./etc/passwd stores basic information about user accounts

Why is /etc/shadow protected?

./etc/shadow contains password hashes and password-related information

What is the purpose of /var/log?

.stores system and application log files

Which file stores the system hostname?

./etc/hostname

Installation Basics

What is a partition?

.A partition is a logically separated section of a physical storage device

What is a filesystem?

.A filesystem is the system Linux uses to organize and manage files and directories on a storage device.

What is the purpose of the swap partition?

.Swap is disk space that Linux can use as additional virtual memory when RAM is under pressure

Why is ext4 commonly used?

.Journaling helps the filesystem recover more safely after events such as a sudden power loss